Dr. NITIN

Designation: Assistant Professor (Mathematics)
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Personal Information:

Nationality: Indian
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Date of Birth: 23/07/1995
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Marital Status: Married
Languages - Hindi: Native; English: Fluent
Gender – Male
Category - General

Research Interest:

- Extreme and high-impact weather events, and associated climatology;
- Urbanization, meteorology, climatology, and associated socio-economic impacts
- Numerical Modeling (i.e. WRF, WRFDA)
- Big Data Analysis
- Climate Change
- Applications of modern techniques and technologies, viz., numerical/ lagrangian/ stochastic/ statistical modeling, data assimilation, remote sensing and GIS techniques, ML/DL in atmospheric studies

Technical Skills:

- **Programming Language:** Python (Expert)
- **Software:** GrADS, ArGIS, HDFView, Panoply, NCView, CDO
- **Technical Proficiency:** Model runs on HPC, SLURM/PBS handling, WRF installation, Parallel computing, Automation & scheduling
- **Machine Learning** (Intermediate)

Exam Qualified:

- National Eligibility Test (NET – Mathematical Sciences): 2019, 2022
- GATE Mathematics: 2022

Education:

- **PhD in Applied Mathematics** (2020-2026) (Defended: 31/01/2026)

  Gautam Buddha University, Greater Noida, India **(8.29CGPA)**

  Thesis Title: Diagnostic and Numerical Modeling Study of Heavy Precipitation Events over Indian Region

  Guide Name: Dr. Sushil Kumar

- **M.Sc. – Maths** (2016-2018)
  DAV College, Panjab University, Chandigarh **(59.35%)**

- **B.Sc. – Physics, Chemistry, Maths** (2013-2016)
  Govt. PG College Jind, Kurukshetra University, Haryana **(77.50%)**

- **HSC – 12th Grade** (2012-2013)
  Haryana Board of Secondary Education (HBSE) **(86.00%)**

- **SSC – 10th Grade** (2010-2011)
  Haryana Board of Secondary Education (HBSE) **(92.80%)**

Collaborations:

- Strong team player with effective collaboration skills
- **NCMRWF:** Worked with Dr. Ashish Routray (Scientist-F) on collaborative research, gaining hands-on experience in model execution.
- **India Meteorological Department (IMD):** Collaborated with Dr. R.K. Giri (Scientist-F) and Dr. D.R. Pattanaik (Scientist-G) on HPC-based modelling experiments; gained exposure to IMD's operational workflows and learned 3DVAR data assimilation techniques.

Research Publications: (Journal Articles)

1. **Lohan, N.,** Pattanaik, D. R., Kumar, S., & Giri, R. K. (2025). **Diagnostic Study of Monsoon-Linked Rainfall Events: Unraveling the Role of Monsoon Low Pressure Systems across the Indian Subcontinent.** *Earth Systems and Environment*, 1-14. DOI: https://doi.org/10.1007/s41748-025-00706-y
2. **Lohan, N.,** Routray, A., Kumar, R., Mahala, B. K., & Kumar, S. (2025). **Validation and diagnostic study of cloudburst events over the Himalayan region using IMDAA and ERA5.** *Geomatics, Natural Hazards and Risk*, *16*(1), 2446589. DOI: https://doi.org/10.1080/19475705.2024.2446589
3. **Lohan, N.,** Kumar, S., Singh, V., Gupta, R. P., & Tiwari, G. (2025). **Analyzing an Extreme Rainfall Event in Himachal Pradesh, India, to Contribute to Sustainable Development.** *Sustainability*, *17*(5), 2115. DOI: https://doi.org/10.3390/su17052115
4. Singh, N., Vinayak Shukla, M., **Lohan, N.,** Mitra, A. K., & Giri, R. K. (2025). **Evaluation of INSAT-3DR sounder derived wind index using radiosonde observation over India.** *Theoretical and Applied Climatology*, *156*(1), 58. DOI: https://doi.org/10.1007/s00704-024-05284-z
5. Kumar, A., Kumar, S., Kumar, N., **Lohan, N.**, & Nupur, R. (2025). **Estimating the Influence of Cumulus Parametrization Schemes in Simulation of Super Cyclonic Storm 'Amphan' Over the Bay of Bengal Using WRF-ARW Model.** *Indian Journal of Pure & Applied Physics (IJPAP*), 63(10). DOI: https://doi.org/10.56042/ijpap.v63i10.11386
6. Kumar, A., Kumar, S., Kumar, N., **Lohan, N.,** & Bhatla, R. (2024). **Comparison of characteristics associated with super cyclonic storm 'Amphan' using numerical model WRF-ARW analysis and ERA5 reanalysis.** *Journal of the Indian Geophysical Union*, *28*(2), 99-107.
7. **Lohan, N**., Kumar, S., Singh, V., & Tiwari, G. (2025). **Role of Microphysics and Convective Parameterization in Simulating Extreme Precipitation Events over the Himalayas: A Multi-Event WRF Analysis.** *Atmospheric Research* (Communicated).
8. **Lohan, N.,** Kumar, S., Mishra, P., Kumar, S., & Giri, R. K. (2023). **A study of the long-term trend of methane emission over the Indian region during 1970–2020.** *VayuMandal,* 49(2), 26–33. Indian Meteorological Society.

Conference Proceedings:

1. **Lohan, N.,** Tiwari, G., Kumar, S., & Routray, A. (2023, March). **On the Analysis of a Very Severe Cyclonic Storm Nilofar Over the Arabian Sea: A Numerical Weather Prediction Model Study.** In *International Conference on Applied and Industrial Mathematics* (pp. 157-169). Singapore: Springer Nature Singapore.
   DOI: https://doi.org/10.1007/978-981-96-3460-6_10
2. Panda, J., Paul, D., Sarkar, A., Mukherjee, A., Bhasi, I., Tom, G., **Lohan N.** & Kumar, S. (2025). **Severe storms in changing climate scenario with a distinctive emphasis on South Asia and India.** In *Severe storms: anatomy, early warning systems and aftermath in changing climate scenarios* (pp. 563-606). Singapore: Springer Nature Singapore.
   DOI: https://doi.org/10.1007/978-981-97-7075-5_22
3. Kumar, N., Kumar, S., Panda, B., Routray, A., Kumari, N., & **Lohan, N.** (2025, January). **Diagnostic Study of Thunderstorms Over Uttar Pradesh, India**. In *2025 International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI)* (pp. 1276-1279). IEEE.
   DOI: https://doi.org/10.1109/IC3ECSBHI63591.2025.10990776

Conference Attended:

1. ***Nitin Lohan**, Sushil Kumar, and Gaurav Tiwari, 2022.* Diagnostic study of cloudburst events over the Himalayan region using Reanalysis Datasets. **TROPMET – 2022**, Bhopal Chapter, IISER Bhopal, India. *(Poster Presentation)*
2. ***Nitin Lohan**, Sushil Kumar, R. K. Giri, and D.R. Pattanaik. 2023.* Role of Monsson Low Pressure System in EREs over Indian Subcontinent. **TROPMET – 2023**, Jaipur Chapter. Birla Institute of Scientific Research, Jaipur, India. *(Oral Presentation)*
3. ***Nitin Lohan**, Sushil Kumar, and Gaurav Tiwari. 2023.* On the Analysis of a Very Severe Cyclonic Storm Nilofar over the Arabian Sea: A numerical weather prediction model study. **ICAIM-2023.** Sharda University, India. *(Oral Presentation)*

Workshop & Training Program:

- Attended a 05 days' **Regional Capacity Building Programme: Series – 3** on **Inspiring the Minds for Disaster Risk Reduction**, organised by *National Institute of Disaster Management, India* from 06 Feb to 10 Feb 2023.
- Successfully completed the online **Induction Training Program on Meteorology** conducted by *Meteorological Training Institute, IMD-Pune* from 22 April to 31 May 2024.
- Attented 02-Day **National Workshop on Leadership for Promoting Ethics and Morality in Research** organised by *Panjab University, Chandigarh* from 05 March – 06 March 2020.

REFERENCES:

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Dr. R. K. Giri

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